



DEPARTMENT OF SOFTWARE ENGINEERING
Semester-V

Lab number : 01

**Lab Title: GitHub Identity, Student Portal Registration, and
Ubuntu Server Installation**

Name: Ayesha Afifa

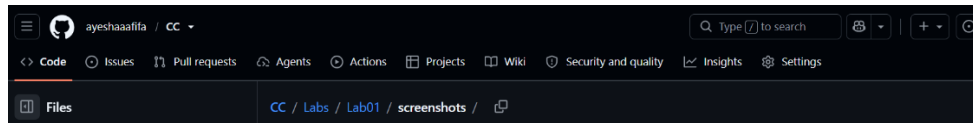
Registration number: FA24B1-SE-001

Course code: BSE-665

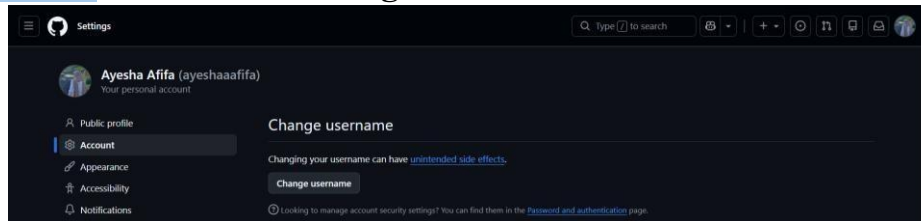
Section: A

GitHub username: ayeshaafifa

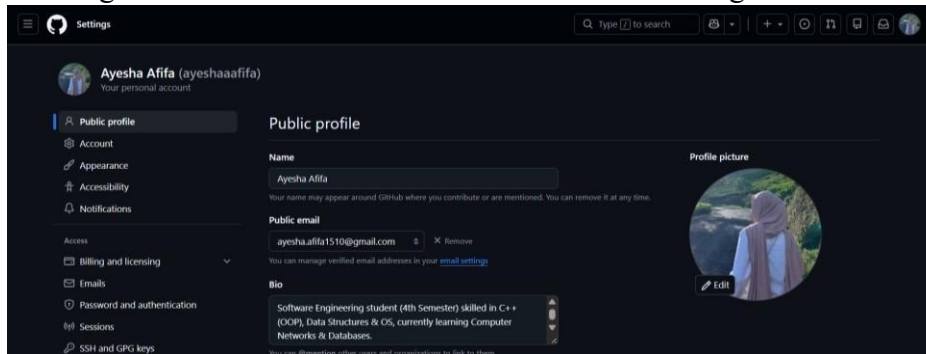
Getting Started



Task 1: Create and Configure Your GitHub Account



Renaming the GitHub account username in Account settings.



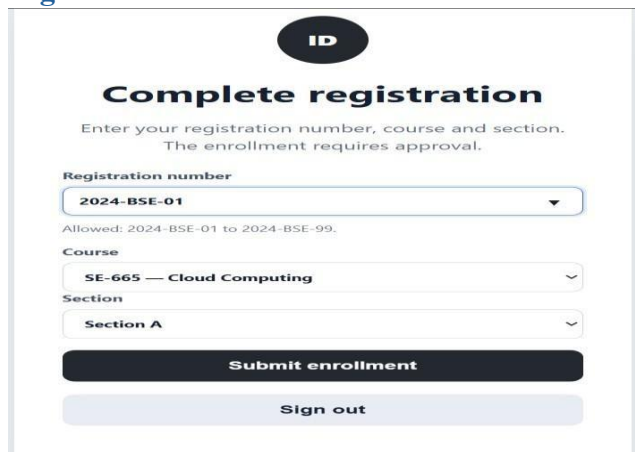
GitHub public profile updated with name, photo, and bio.

Task 2: Register in the Student Marks Portal



Authorizing Firebase GitHub Login to access the account for the Student Marks Portal.

Register:



The registration form is titled "Complete registration" and includes an "ID" icon. It prompts the user to enter their registration number, course, and section. The registration number is set to "2024-BSE-01", the course is "SE-665 — Cloud Computing", and the section is "Section A". There are "Submit enrollment" and "Sign out" buttons.

Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number
2024-BSE-01

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course
SE-665 — Cloud Computing

Section
Section A

Submit enrollment

Sign out

Completing enrollment with registration number, course, and section on the Student Marks Portal.

Enrollment:



The "Course enrollments" page shows the user's name "Ayesha Afifa" and their registration number "2024-BSE-01". It includes a "Sign out" button and a link to "Enroll in another course". The "ENROLLMENT STATUS" section shows "Your courses" with "Cloud Computing (SE-665) Section A" listed as "Waiting for administrator approval" and "Pending approval".

STUDENT MARKS PORTAL

Course enrollments

Sign out

Ayesha Afifa
2024-BSE-01

+ Enroll in another course

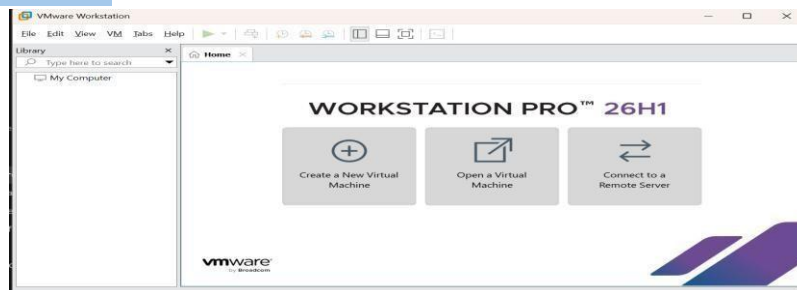
ENROLLMENT STATUS

Your courses

Cloud Computing (SE-665) Section A
Waiting for administrator approval
Pending approval

Enrollment confirmed, status shown as pending administrator approval.

Task 3: Download VMware Workstation Pro and Ubuntu Server



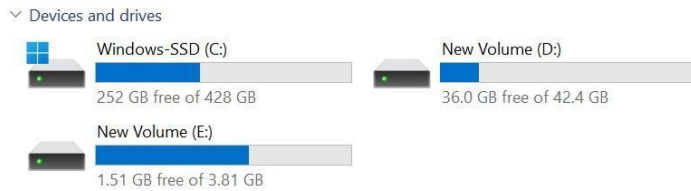
VMware Workstation Pro 26H1 home screen after installation.

Ubuntu iso file:



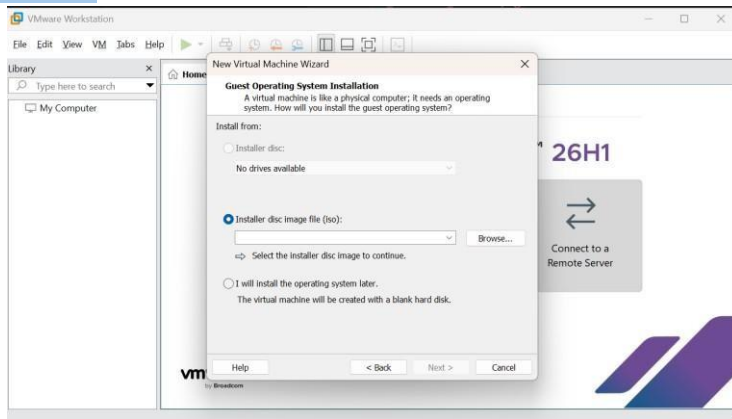
Download history confirming the Ubuntu Server ISO was downloaded.

Available Storage:



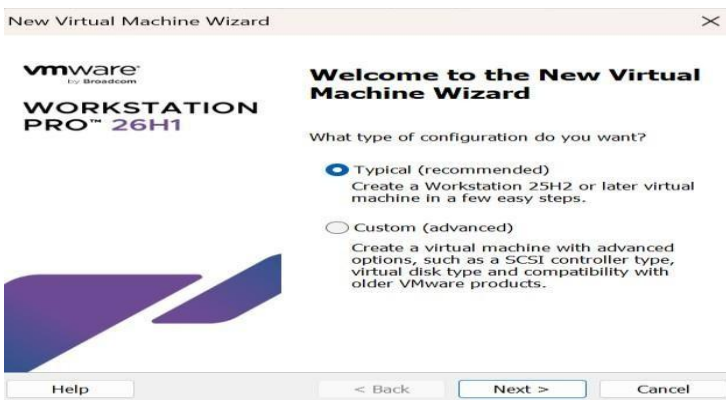
Available disk storage on the host machine before creating the VM.

Task 4: Create the Ubuntu Server Virtual Machine



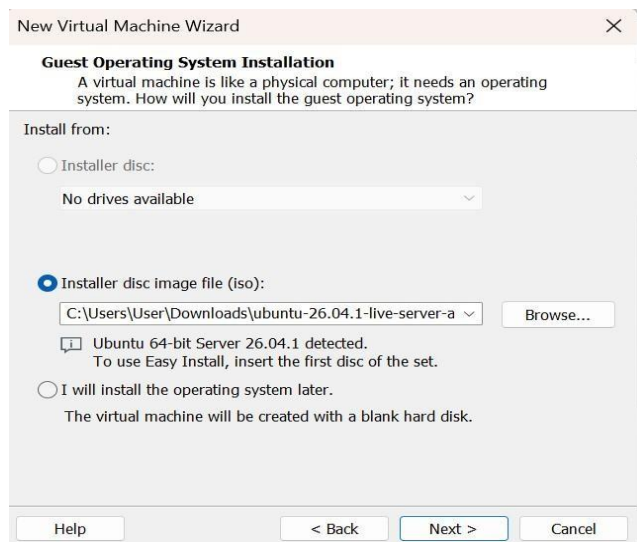
New VM Wizard set to install from an installer disc image file (ISO).

Select typical:



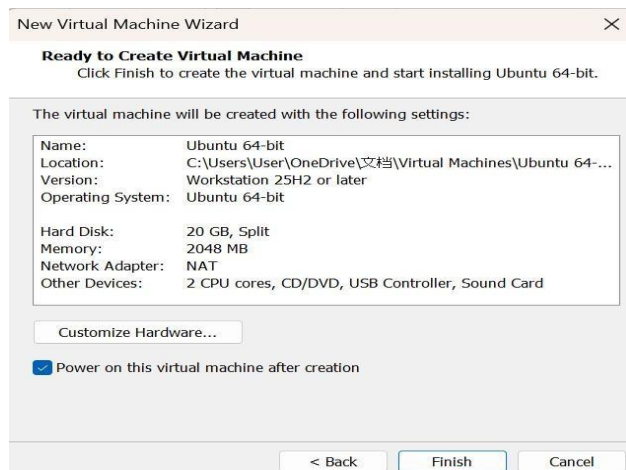
Typical (recommended) configuration selected for the new VM.

Browse ubuntu server file:

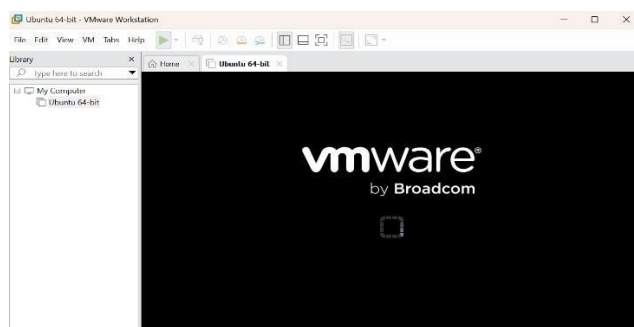


Ubuntu Server ISO selected as the install source; installer detects Ubuntu 64-bit Server 26.04.1.

Configuration Summary

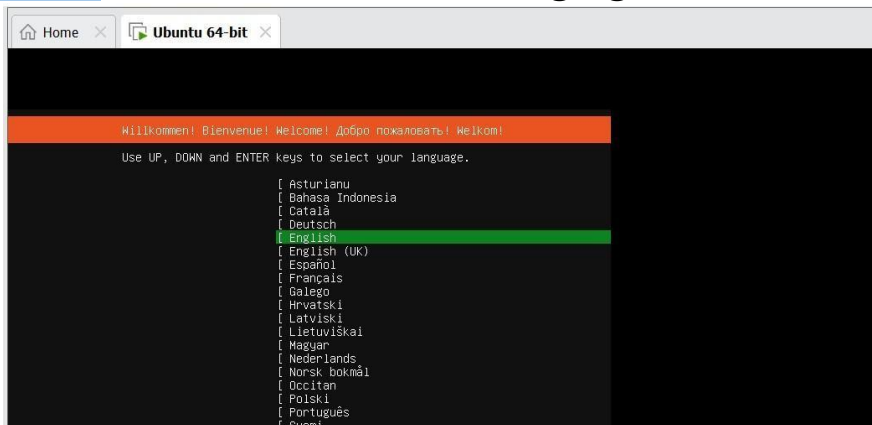


Final configuration summary before creation: name, 20 GB disk, memory, network. **Ubuntu** installer boot waiting



VM powered on, waiting for the Ubuntu Server installer to boot.

Task 5: Choose the Installation Language



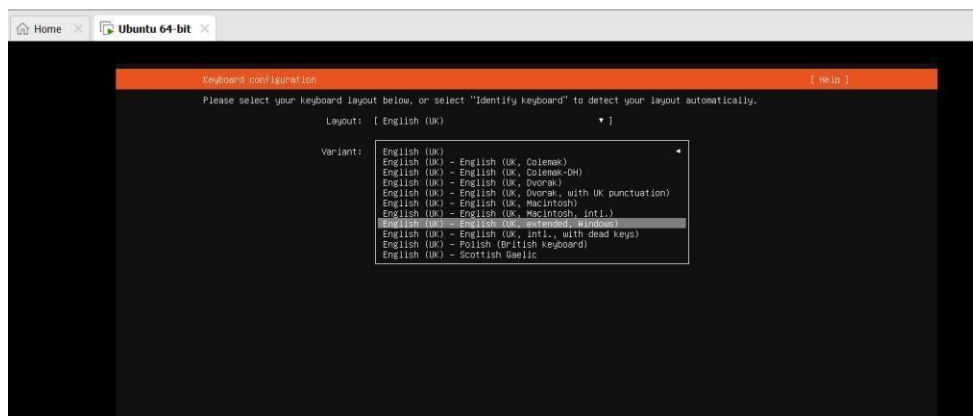
Installer language selection screen with English highlighted.

Task 6: Configure the Keyboard Layout [Layout:](#)



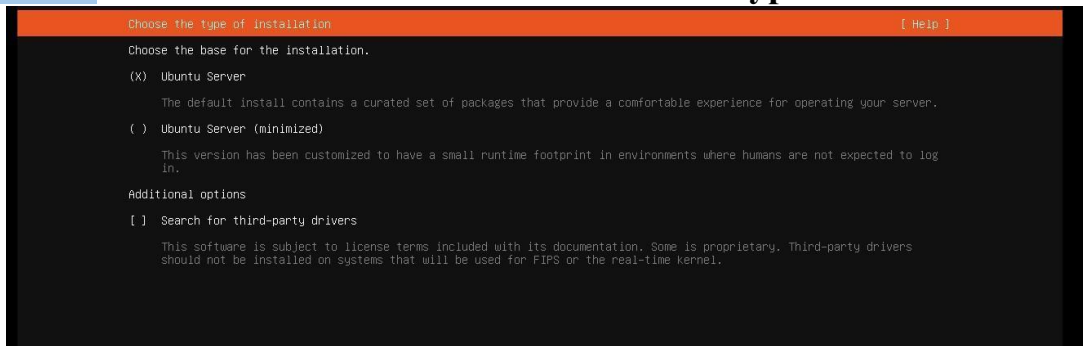
Keyboard layout screen showing English (US) selected.

Variant :



Keyboard variant dropdown expanded with English (UK) options.

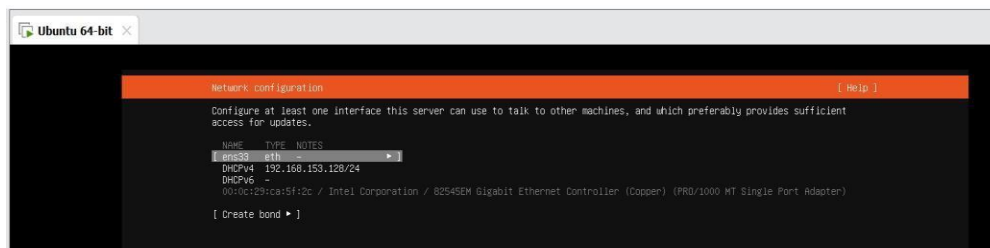
Task 7: Choose the Ubuntu Server Installation Type



Installation type screen with 'Ubuntu Server' selected.

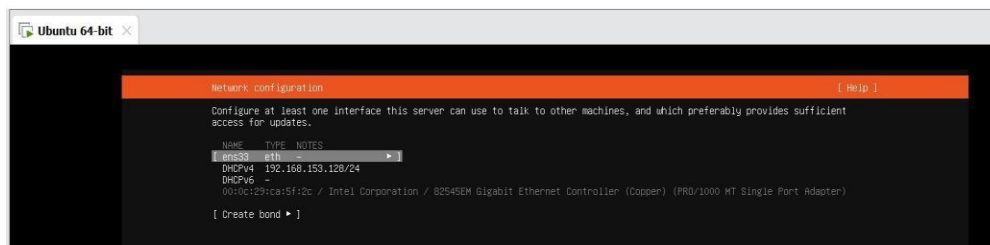
Task 8: Configure the Network

Network Interface:



Network configuration screen showing the detected interface (ens33).

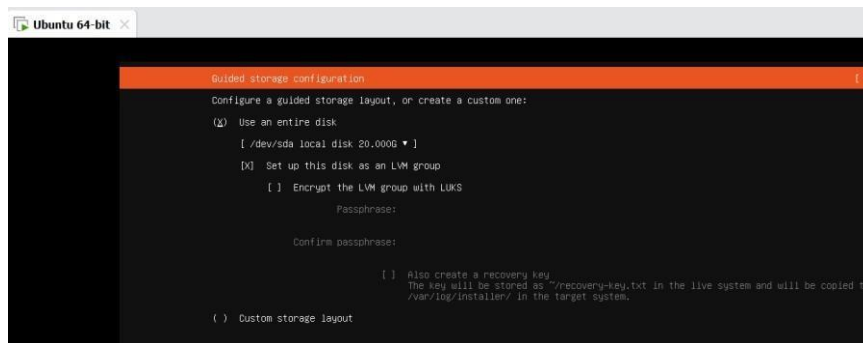
Installer network:



Interface confirmed to have received an IPv4 address via DHCP.

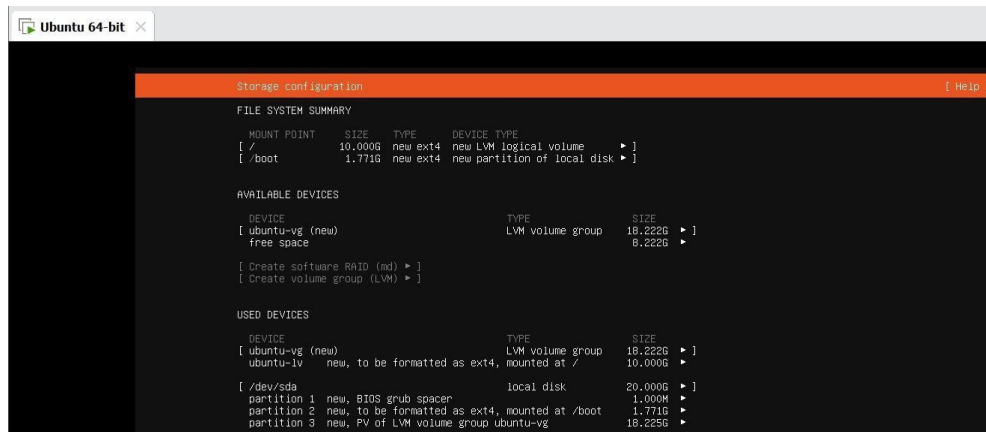
Task 9: Configure Storage

Guided Storage



Guided storage: 'Use an entire disk' selected on the 20 GB VMware disk.

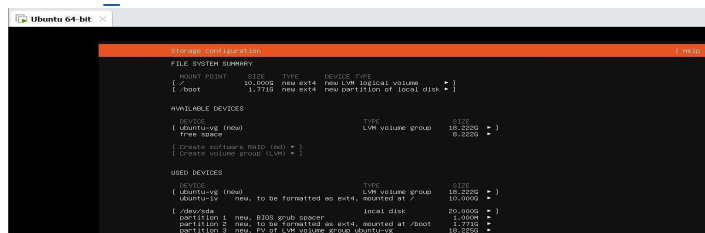
Storage Configuration:



Proposed file system layout (root and boot) on the virtual disk.

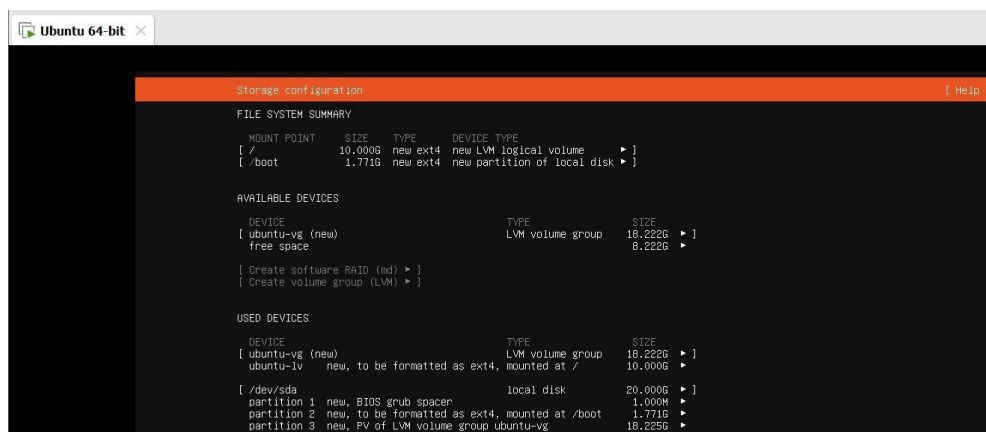
Task 10: Select the Installation Disk

Selected_disk



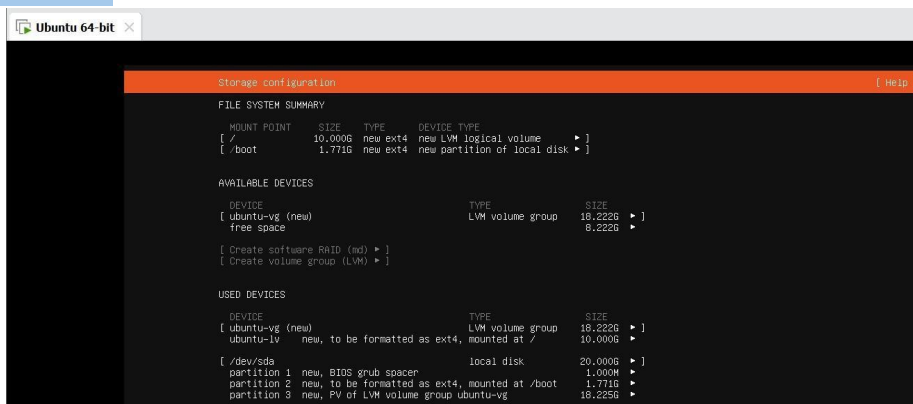
Selected installation disk confirmed as the VMware virtual disk.

Disk_details:



Disk details showing device name, partitions, and sizes.

Task 11: Review the Partitions



Complete partition layout reviewed before confirming changes.

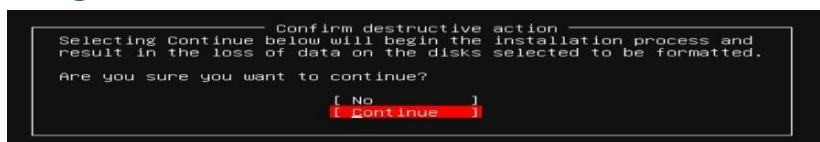
Task 12: Confirm the Storage Changes

Storage Warning



Destructive-action warning shown, with 'No' highlighted by default.

Storage Confirmed:



Selection moved to 'Continue' to confirm the storage changes.

Task 13: Configure the Ubuntu Profile

Actual Name:



Profile configuration: actual name 'Ayesha Afifa' entered.

Server name:

Server name set to 'ubuntu' during profile configuration.

Username:

Login username 'ayeshaaafifa' chosen during profile configuration.

Review profile:

Completed profile with name, server name, username, and password set.

Task 14: Complete the Software Installation

Software installation in progress, showing the installer's package log.

Task 15: Reboot and Verify the Ubuntu Identity

Installation complete:

```
Installation complete! [ Help ]

curtin command extract
  acquiring and extracting image from cp:///tmp/tmptculf032/mount
curtin command in-target
  executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
curtin command curthooks
  configuring apt
  configuring apt
  installing missing packages
  configuring lscsi service
  configuring raid (mdadm) service
  configuring NVMe over TCP
  installing kernel
  setting up swap
  applying networking config
  writing etc/fstab
  configuring multipath
  updating packages on target system
  configuring pollinate user-agent on target
  configuring kernel crash dumps settings
  final kernel configuration
  configuring target system bootloader
  installing grub to target devices
  copying metadata from /cdrom
final system configuration
curtin command in-target
  calculating extra packages to install
  installing openssh-server
  retrieving openssh-server
curtin command system-install
  unpacking openssh-server
curtin command system-install
  configuring cloud-init
  downloading and installing security updates
curtin command in-target
  restoring apt configuration
curtin command in-target
subiquity/Late/run_builtins:
subiquity/Late/run_user_supplied:
```

Installation complete, ready to reboot into Ubuntu Server.

Ubuntu login screen:

```
ubuntu login:
ubuntu login: ayeshaafifa
Password: _
```

Ubuntu login prompt after reboot with username 'ayeshaafifa'.

Ubuntu terminal login:

```
ubuntu login:
ubuntu login: ayeshaafifa
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sun Sep 13 09:41:30 AM UTC 2026

System load:  0.0      Processes:      216
Usage of /:   50.5% of 9.75GB   Users logged in:  0
Memory usage: 21%      IPv4 address for ens33: 192.168.153.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ayeshaafifa@ubuntu:~$
```

Terminal login successful, showing system info and IPv4 for ens33.

Ubuntu identity verified:

```
ayeshaafifa@ubuntu:~$ _
```

Terminal prompt confirming the logged-in Ubuntu identity.

Task 16: Find the Ubuntu Server IP Address

Ip addr command

```
ayeshaaafifa@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:ca:5f:2c brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29ca5f2c
    inet 192.168.153.129/24 metric 100 brd 192.168.153.255 scope global dynamic ens33
        valid_lft 970sec preferred_lft 970sec
    inet6 fe80::20c:29ff:eca:5f2c/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
ayeshaaafifa@ubuntu:~$
```

Output of 'ip addr' showing network interface details for ens33. **Ubuntu**

ip address

```
ayeshaaafifa@ubuntu:~$ ip address
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:ca:5f:2c brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29ca5f2c
    inet 192.168.153.129/24 metric 100 brd 192.168.153.255 scope global dynamic ens33
        valid_lft 970sec preferred_lft 970sec
    inet6 fe80::20c:29ff:eca:5f2c/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
ayeshaaafifa@ubuntu:~$
```

Close-up of the assigned IPv4 address 192.168.153.129.

Task 17: Connect to Ubuntu Server from Windows

Windows ssh command

```
Command Prompt - ssh ayesi
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh ayeshaafifa@192.168.153.129
The authenticity of host '192.168.153.129 (192.168.153.129)' can't be established.
ED25519 key fingerprint is SHA256:aUM/s3XiaoR5wggnc8CTR97T/BERhtjbwQj4T88Nee8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

SSH connection started from Windows CMD; host key fingerprint prompt shown.

Ssh_fingerprint:

```
ayeshaaafifa@ubuntu: ~
The authenticity of host '192.168.153.129 (192.168.153.129)' can't be established.
ED25519 key fingerprint is SHA256:aUM/s3XiaoR5wggnc8CTR97T/BERhtjbwQj4T88Nee8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.153.129' (ED25519) to the list of known hosts.
ayeshaaafifa@192.168.153.129's password:
```

Fingerprint accepted ('yes'); password prompt shown for SSH login.

ssh login:

```
ayeshaaafifa@192.168.153.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Sep 13 05:49:29 PM UTC 2026

System load: 0.03          Processes:      223
Usage of /:  50.6% of 9.75GB Users logged in:  0
Memory usage: 23%         IPv4 address for ens33: 192.168.153.129
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

ayeshaaafifa@ubuntu:~$ |
```

Successful SSH login from Windows into the
Ubuntu Server VM.

ssh identity:

```
Last login: Sun Sep 13 17:51:21 2026 from 192.168.153.1
ayeshaaafifa@ubuntu:~$ |
```

Terminal prompt confirming the Ubuntu Server identity over SSH.
